The example of training data sets consists of three coefficients with 10 values each:

```
aa<-c(1, 1,  -3,  1, -5, 2,2,2,1,1)
bb<-c(5, -3,  -1,  10, 7, -7,1,1,-4,-25)
cc<-c(6, -10, -1,  24, 9, 3,-4,4,-21,156)
```

Data preprocessing

To discard equations with zero as coefficients of x^2, use the following code:

```
k <- which(aa != 0)
aa <-aa[k]
bb <-bb[k]
cc <-cc[k]
```

To accept only those coefficients for which the discriminant is zero or more, use the code given below:

```
disc <-(bb*bb-4*aa*cc)
k <- which(disc >= 0)
aa <-aa[k]
bb <-bb[k]
cc <-cc[k]
a <-  as.data.frame(aa)                # a,b,c vectors are converted to data frame
b <-  as.data.frame(bb)
c <-  as.data.frame(cc)
```

Calculate the roots of valid equations using conventional formulae, for training and verification of the machine's results at a later stage.

```
r1 <- (-b + sqrt(b*b-4*a*c))/(2*a)  # r1 and r2 roots of each equations
r2 <- (-b - sqrt(b*b-4*a*c))/(2*a)
```

After getting all the coefficients and roots of the equations, concatenate them columnwise to form the input-output data sets of a neural network.

```
trainingdata <- cbind(a,b,c,r1,r2)
```

Since this is a simple problem, the network is configured with three nodes in the input-layer, one hidden-layer with seven nodes and a two-node output-layer.

R function *neuralnet()* requires input-output data in a proper format. The format of formulation procedure is somewhat tricky and requires attention. The right hand side of the formula consists of two roots and the left side includes three coefficients a, b and c. The inclusion is represented by + signs.

```
colnames(trainingdata) <- c(“a”,”b”,”c”,”r1”,”r2”)
```